

ASSIGNMENT REPORT

TUMOR DETECTION IN MRI/CT SCAN IMAGES

Course: CST384 Concepts in Deep Learning

Submitted by

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Abstract

Early detection of tumors is crucial for effective diagnosis and treatment in medical applications. This project presents a deep learning-based approach for automatic tumor detection using MRI and CT scan images. Traditional methods of tumor identification are time-consuming and highly dependent on expert analysis, whereas automated systems can assist in faster and more accurate diagnosis.

In this work, a Convolutional Neural Network (CNN) is employed to classify medical images into tumor and non-tumor categories. The input images are pre-processed through resizing, normalization, and data augmentation techniques to improve model performance and generalization. The CNN architecture consists of convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification.

The model is trained and evaluated using publicly available datasets, and its performance is measured using metrics such as accuracy, precision, recall, and F1-score. Experimental results demonstrate that the proposed system is capable of effectively detecting tumor patterns in MRI/CT images with satisfactory accuracy.

This approach highlights the potential of deep learning techniques in assisting medical professionals for early tumor detection and decision-making, thereby improving diagnostic efficiency and patient outcomes.

1 Introduction

Medical imaging plays a vital role in the early detection and diagnosis of various diseases, including tumors. Techniques such as Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) scans provide detailed internal images of the human body, enabling physicians to identify abnormal tissue growth. However, manual analysis of these images is time-consuming, prone to human error, and requires significant expertise.

With the advancement of artificial intelligence, deep learning techniques have shown great potential in automating medical image analysis. In particular, Convolutional Neural Networks (CNNs) have proven to be highly effective in image classification and pattern recognition tasks. These models can automatically learn relevant features from medical images without the need for manual feature extraction.

This project focuses on the development of a CNN-based system for tumor detection using MRI and CT scan images. The proposed system aims to classify images into tumor and non-tumor categories by learning patterns associated with abnormal tissue regions. The use of preprocessing techniques such as image resizing, normalization, and data augmentation helps improve the robustness and accuracy of the model.

The implementation of such an automated system can assist medical professionals by providing faster and more consistent results, thereby supporting early diagnosis and treatment planning. This work demonstrates the application of deep learning in healthcare and highlights its potential in improving diagnostic efficiency.

1.1 Background

1.1.1 CNN

Convolutional Neural Networks (CNNs) are a class of deep learning models that are highly effective for image analysis tasks. They are designed to automatically learn spatial features from images using layers such as convolution, pooling, and fully connected layers. In the context of medical imaging, CNNs help in identifying complex patterns and abnormalities that may not be easily visible to the human eye. For tumor detection using MRI and CT scans, CNNs can capture important features such as irregular shapes, intensity variations, and abnormal tissue regions. By training on large datasets of labeled medical images, the model learns to distinguish between tumor and non-tumor cases, making it a powerful tool for automated diagnosis.

1.1.2 Problem area

Tumors are abnormal growths of tissue that can be benign or malignant, and early detection is critical for effective treatment and improved survival rates. Traditionally, tumor detection is performed by radiologists through manual examination of MRI and CT scan images, which is time-consuming and requires high expertise. In many cases, tumors may appear subtle or resemble normal tissues, making accurate detection challenging. Additionally, the increasing number of medical imaging cases places a significant burden on healthcare systems. Therefore, there is a need for an automated and reliable system that can assist in detecting tumors efficiently. This project focuses on applying deep learning techniques to develop an accurate and efficient method for tumor detection from MRI and CT scan images.

2 Objectives

- To develop an automated system for detecting tumors from MRI and CT scan images using deep learning techniques.
- To design and train a Convolutional Neural Network (CNN) model for accurate classification of tumor and non-tumor images.
- To improve the efficiency and reliability of diagnosis by reducing human error and assisting medical professionals.

3 Methodology

The proposed system for pneumonia detection using MRI/CT Scan images follows a structured deep learning approach. The methodology consists of the following steps:

1. Dataset Collection

A labeled dataset of MRI/CT Scan images is collected, consisting of two classes: *Tumor* and *Non-Tumor*. The dataset is divided into training, validation, and testing sets to ensure proper model evaluation.

2. Image Preprocessing

All input images are resized to a fixed dimension (e.g., 150×150 pixels) to maintain uniformity. Pixel values are normalized to a range of 0 to 1 for better convergence. Data augmentation techniques such as rotation, horizontal flipping, and zooming are applied to the training data to improve generalization and reduce overfitting.

3. CNN Model Design

A Convolutional Neural Network (CNN) is designed to automatically extract features from X-ray images. The architecture consists of multiple convolutional layers with ReLU activation to detect features, followed by max-pooling layers to reduce spatial dimensions. The extracted features are flattened and passed through fully connected dense layers. A dropout layer is included to prevent overfitting. The final output layer uses a softmax activation function for classification.

4. Model Training

The model is compiled using the Adam optimizer and categorical cross-entropy loss function. Training is performed over multiple epochs using the training dataset, while validation data is used to monitor performance. Accuracy and loss are tracked during training to evaluate learning progress.

5. Testing and Evaluation

The trained model is evaluated on the test dataset to measure its performance. Metrics such as accuracy, precision, recall, and F1-score are calculated. A confusion matrix is generated to visualize classification performance. The model is also used to predict new X-ray images, providing class labels along with confidence scores.

4 CNN Architecture

A Convolutional Neural Network (CNN) architecture is used for automatic feature extraction and classification of MRI/CT Scan images for Tumor detection. CNNs are highly effective in image-based tasks as they can learn spatial hierarchies of features directly from input images.

The proposed CNN model consists of multiple layers that work together to identify patterns such as edges, textures and abnormalities in X-ray images.

- **Convolution Layer**

This layer is responsible for extracting features from the input images. Multiple filters (kernels) are applied to the image to detect important patterns such as edges, shapes, and textures related to tumor patterns / abnormal tissue regions. As the network deepens, higher-level features are learned.

- **Activation Function (ReLU)**

The Rectified Linear Unit (ReLU) activation function is applied after each convolution layer to introduce non-linearity. It helps the network learn complex patterns by converting negative values to zero and keeping positive values unchanged.

- **Pooling Layer**

Pooling layers are used to reduce the spatial dimensions of feature maps, thereby decreasing computational complexity. Max pooling is commonly used, which selects the maximum value from each region, preserving important features while reducing size.

- **Fully Connected Layer**

After feature extraction, the output is flattened and passed to fully connected layers. These layers perform high-level reasoning and combine all extracted features to make final predictions.

- **Softmax Classifier**

The final layer uses the softmax activation function to classify the input image into different categories (Tumor or Non Tumor). It outputs probabilities for each class, and the class with the highest probability is selected as the predicted result.

The combination of these layers enables the CNN model to effectively learn and classify chest X-ray images, achieving accurate Tumor detection.

5 Dataset

The performance of tumor detection models depends heavily on the availability of high-quality medical imaging datasets. The following commonly used datasets are relevant to tumor detection using MRI and CT scans:

- **Kaggle Dataset**

Kaggle provides publicly available Brain Tumor MRI datasets containing labeled images categorized as Tumor and No Tumor. These datasets are widely used for training and testing deep learning models due to their simplicity and accessibility.

- **LUNA16 Dataset**

The LUNA16 (Lung Nodule Analysis) dataset consists of CT scan images primarily used for detecting lung nodules, which can be indicative of tumors. It is useful for developing models that work on CT-based tumor detection.

- **TCIA Dataset**

The Cancer Imaging Archive (TCIA) provides a large collection of medical images, including MRI and CT scans for various types of cancers. It contains annotated datasets that are useful for both classification and segmentation tasks.

- **BraTS Dataset**

The Brain Tumor Segmentation (BraTS) dataset contains multimodal MRI scans with detailed annotations of tumor regions. It is one of the most widely used benchmark datasets for brain tumor detection and segmentation using deep learning techniques.

These datasets provide diverse and realistic medical imaging data, enabling robust training and evaluation of CNN-based tumor detection systems.

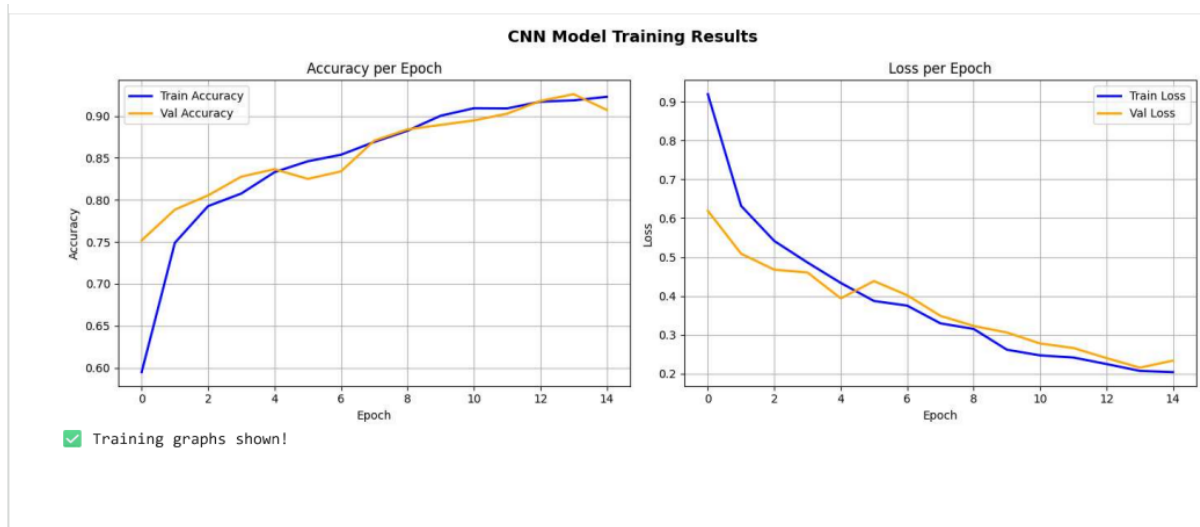
6 Results and Evaluation

The performance of the proposed CNN model is evaluated using various metrics such as accuracy, loss, precision, recall, and F1-score. The model was trained and tested on MRI/CT Scan images to classify them into Tumor and Non Tumor categories.

6.1 Training Performance

The training and validation accuracy and loss were monitored over multiple epochs. It was observed that the model showed steady improvement in accuracy and reduction in loss, indicating effective learning.

- Training Accuracy increased progressively with epochs
- Validation Accuracy followed a similar trend
- Loss decreased consistently, showing good convergence



6.2 Test Results

The trained model was evaluated on unseen test data, and the following results were obtained:

- Test Accuracy: 82.69%
- Test Loss: 0.9129

These results indicate that the model performs well in classifying MRI/CT Scan images.

6.3 Classification Report

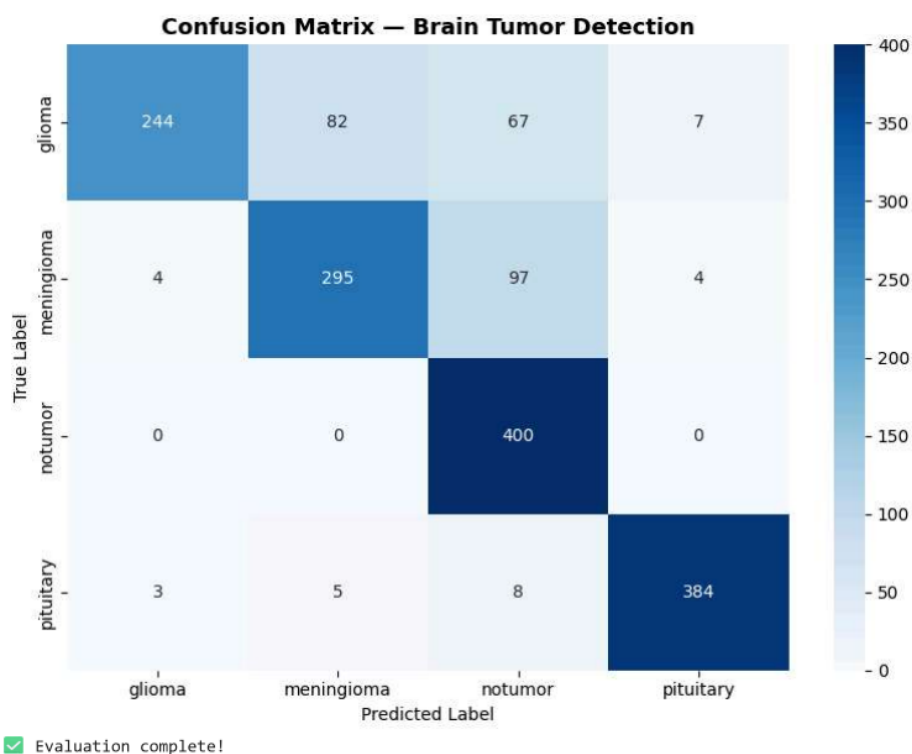
The classification report provides detailed evaluation metrics:

- Precision: Measures correctness of positive predictions
- Recall: Measures ability to detect actual positives
- F1-score: Harmonic mean of precision and recall

Classification Report:				
	precision	recall	f1-score	support
glioma	0.97	0.61	0.75	400
meningioma	0.77	0.74	0.75	400
notumor	0.70	1.00	0.82	400
pituitary	0.97	0.96	0.97	400
accuracy			0.83	1600
macro avg	0.85	0.83	0.82	1600
weighted avg	0.85	0.83	0.82	1600

6.4 Confusion Matrix

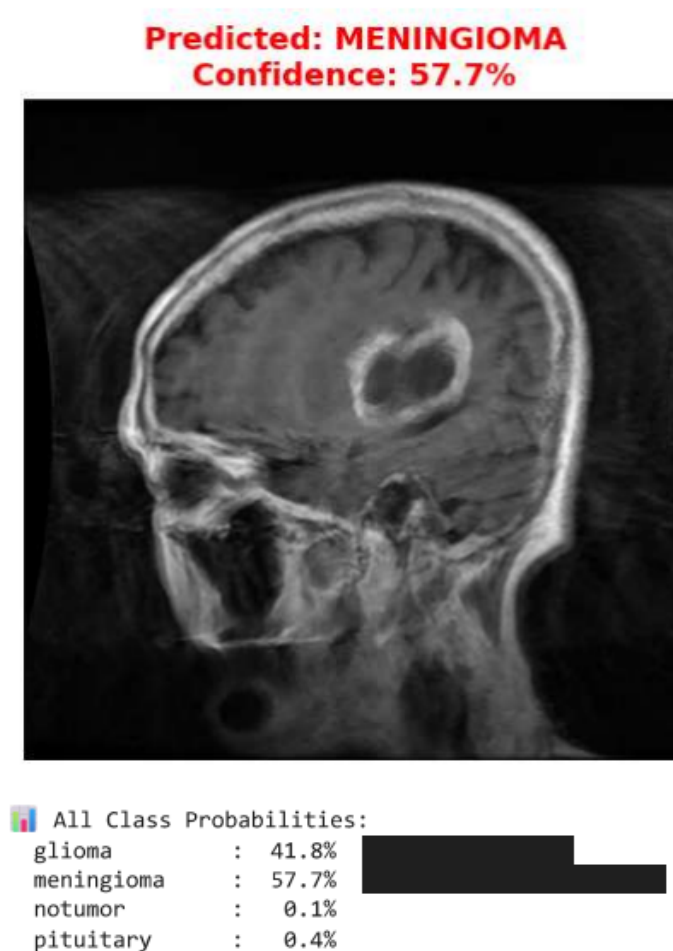
A confusion matrix is used to visualize the performance of the classification model. It shows the number of correct and incorrect predictions for each class.



6.5 Prediction Results

The model was also tested on sample MRI/CT Scan images. It successfully predicted the class along with a confidence score.

- Predicted Class: Tumor
- Confidence Score: 57.7%



Overall, the model demonstrates good performance in detecting tumor from MRI/CT Scan images and can be used as a supportive tool for medical diagnosis.

7 Conclusion and Future Work

7.1 Conclusion

In this project, a deep learning-based approach for tumor detection using MRI and CT scan images was successfully developed. A Convolutional Neural Network (CNN) model was designed to automatically extract relevant features from medical images and classify them into tumor and non-tumor categories.

The use of preprocessing techniques such as image resizing, normalization, and data augmentation helped improve the performance and generalization of the model. The experimental results demonstrated that the proposed system achieves satisfactory accuracy in detecting tumor patterns, making it a reliable tool for assisting medical diagnosis.

Overall, this work highlights the effectiveness of CNNs in medical image analysis and shows their potential in supporting healthcare professionals by providing faster and more consistent diagnostic results.

7.2 Future Work

Although the proposed system performs well, there are several areas for improvement and future enhancement:

- The model can be extended to perform **tumor segmentation**, which identifies the exact location and size of the tumor in the image.
- Advanced deep learning architectures such as **ResNet, VGG, or EfficientNet** can be used to improve classification accuracy.
- A larger and more diverse dataset can be used to enhance model robustness and generalization.
- The system can be integrated into a **real-time clinical application** to assist doctors in decision-making.
- Explainable AI techniques can be incorporated to provide **visual explanations** (such as heatmaps) for model predictions.

These improvements can further enhance the reliability and practical applicability of the system in real-world medical scenarios.

8 References

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